

Aditya Kulkarni

adityakulkarnius@gmail.com | linkedin.com/in/aditya-kulkarni-355b81217 | kulkarniaditya.com | Raleigh, NC

PROFESSIONAL SUMMARY

Systems engineer with hands-on experience in Unix/Linux kernel programming, distributed systems, and containerized application delivery. Engineered a virtual memory system in C for Xinu OS -- 8+ kernel system calls, page fault handler, Second-Chance replacement algorithm -- reducing page faults by 25%. Built Kafka-driven distributed pipelines handling 2,000 async requests per second with Temporal fault-tolerant orchestration; deployed 4+ production systems with Docker across CI/CD pipelines. Python and Java backend across production workloads. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Unix/Linux Systems

Docker + Kubernetes

Python + Java

Distributed Systems

Apache Kafka

Prometheus + Grafana

Cloud-Native (GCP + AWS)

CI/CD + GitHub Actions

WORK EXPERIENCE

NC State University Apr 2025 - Present

Graduate Research Assistant

- Deployed a cloud-native genomic data platform on Google Cloud Run using containerized MERN stack services; enabled interactive exploration of 500GB strawberry and grape datasets for 100+ researchers worldwide with Grafana performance monitoring
- Applied hyperparameter tuning and time-series analysis to ML models, improving accuracy by 12%; maintained reliable distributed data pipelines supporting continuous research workloads

Isomer AI May 2025 - Present

Product Engineer Intern

- Shipped 10+ backend features across 7 production deployments on a containerized TypeScript/Node.js + PostgreSQL platform; managed full SDLC end-to-end with Docker, GitHub Actions CI/CD, and Sentry observability -- 394 commits
- Built Temporal.io workflow orchestration ensuring fault-tolerant at-least-once delivery for AI agent pipelines spanning 3 LLM providers and 8+ models; integrated real-time event pipelines (Pusher) and Auth0 RBAC

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Automated AWS and GCP deployments using Terraform and integrated Grafana monitoring dashboards; built 5+ AI-driven workflows achieving 25% accuracy improvement in health claim processing

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Implemented distributed data pipelines with Hadoop + Spark for batch processing and integrated Kafka streaming for real-time ingestion, cutting ticket-processing latency by 20%; managed end-to-end SDLC for a large-scale data system reducing processing time by 15%

PROJECTS

Demand Paging -- Xinu OS Unix/Linux + Systems

Engineered a complete virtual memory system in C for Xinu OS: implemented Second-Chance page replacement algorithm, added 7+ kernel system calls, built an inverted page table, and wrote a custom page fault handler -- reducing page faults by 25% and improving memory performance by 15%. Work spans kernel-level resource management and low-level system call design, directly analogous to container runtime and resource utilization challenges.
Stack: C, Xinu OS, x86 virtual memory, kernel systems programming

AlgoPulse -- Agentic AI Platform Distributed + Kafka

Architected an event-driven AI orchestration platform; identified API-polling bottleneck under load and re-architected to Supabase Edge Functions for streaming AI reasoning loops. Scaled with Temporal.io and Kafka for fault-tolerant state synchronization across worker nodes -- 2,000 async inference requests per second without state loss under load.

Stack: TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka, Docker

VectorLift -- Semantic Search and Ranking Engine Python + Docker + Kafka

Built a containerized semantic search system on MS MARCO: BM25 (Elasticsearch), dense bi-encoder retrieval (sentence-transformers), and cross-encoder re-ranker (MiniLM) across 6 pipeline configurations. Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; served via FastAPI with per-stage latency tracking in Prometheus + Grafana. Full Docker Compose orchestration.

Stack: Python, PyTorch, Elasticsearch, Kafka, FastAPI, Qdrant, Prometheus, Grafana, Docker Compose, Redis

Inventory Reservation Service Java + Concurrency

Transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across concurrent requests; Testcontainers integration tests verifying exactly-one-success under simultaneous load. Multi-stage Docker build, Flyway migrations, GitHub Actions CI.

Stack: Java 21, Spring Boot 3.5, PostgreSQL, Docker, Flyway, Testcontainers, GitHub Actions

EDUCATION

North Carolina State University -- Master of Science, Computer Science Aug 2024 - May 2026

Courses: Operating Systems, Artificial Intelligence, Distributed Systems, Data Science, Algorithms, Software Engineering

Mumbai University (DJSCE) -- Bachelor of Technology, Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | IEEE-published: NLP document-to-graph pipeline (T5 transformer, 1,500+ PDFs processed)

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL Systems: Linux/Unix, kernel programming, virtual memory, system calls, process scheduling

Containers + Infra: Docker, Kubernetes, Terraform, Docker Compose, GitHub Actions, CI/CD

Distributed: Apache Kafka, Temporal.io, Spark, Hadoop, Redis, event-driven architecture Observability: Prometheus, Grafana, Sentry, OpenTelemetry

Cloud: AWS, GCP (Cloud Run), Google Cloud AI/ML: PyTorch, sentence-transformers, Elasticsearch, FAISS, OpenAI/Anthropic/Gemini APIs